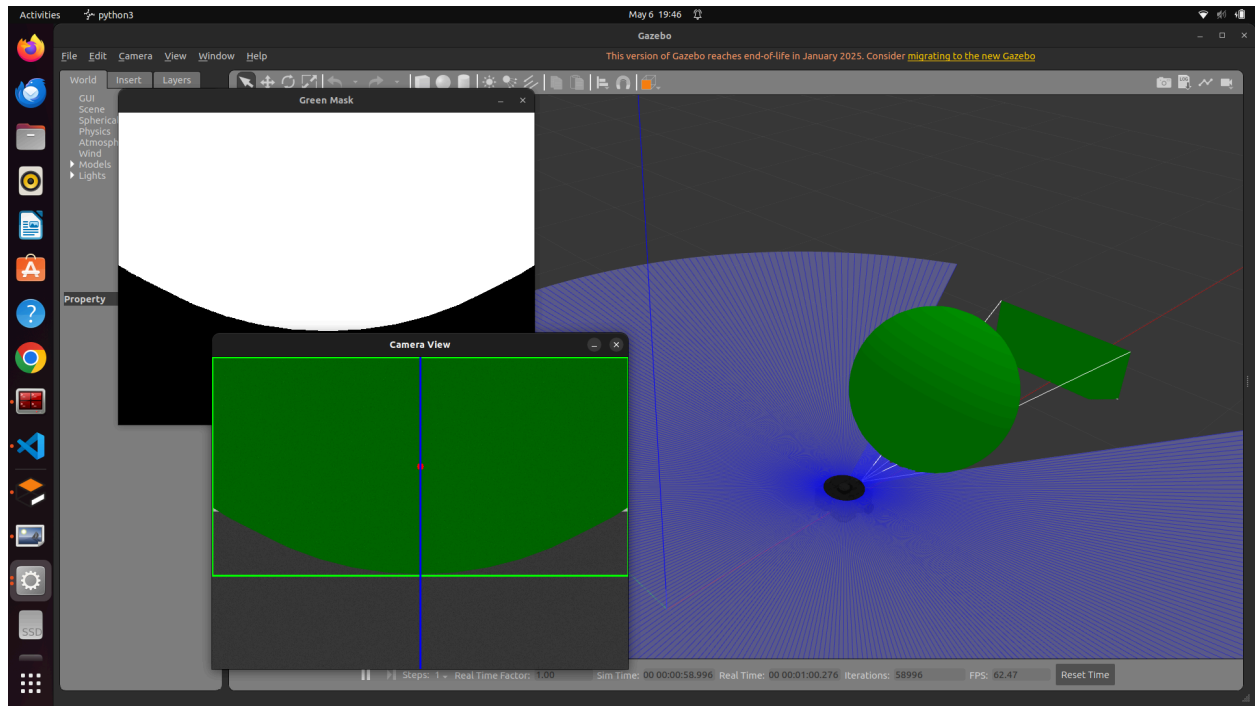
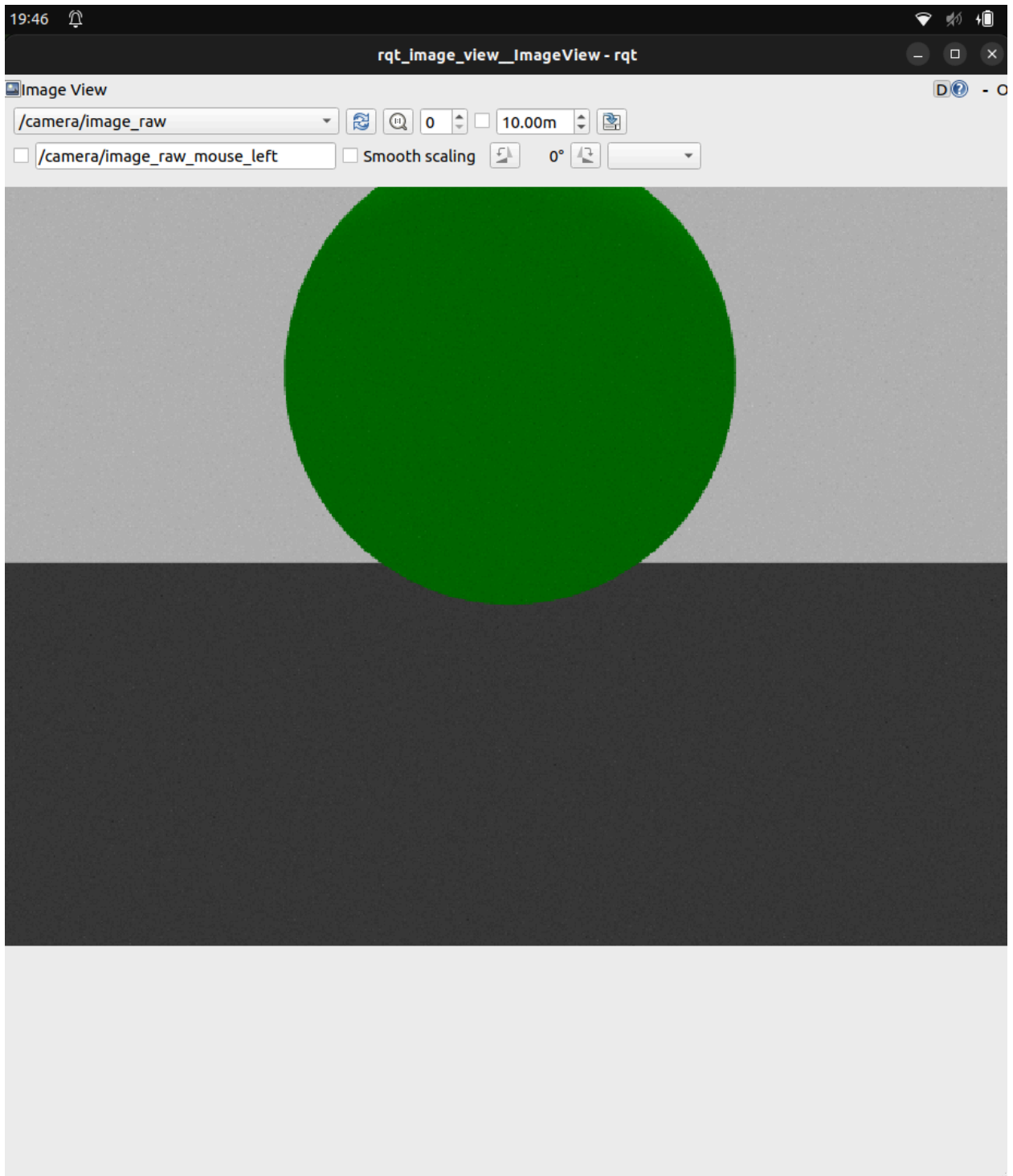


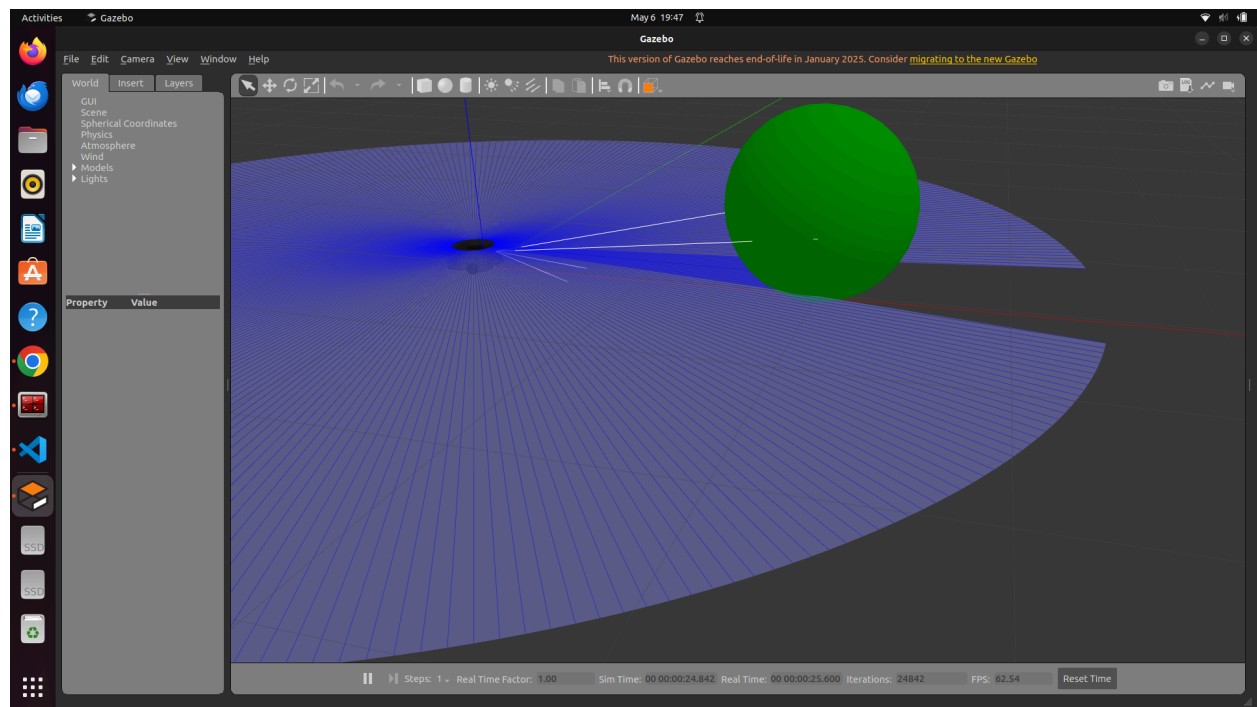




```

priyanshu@priyanshu-hp-pavilion:~/ros2_ws$ ros2 launch tb3_custom_world green_sphere_world.launch.py
[INFO] [launch]: All log files can be found below /home/priyanshu/.ros/log/2026-05-06-19-47-06-249226-priyanshu-hp-pavilion-17149
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [gzserver-1]: process started with pid [17150]
[INFO] [gzclient-2]: process started with pid [17152]
[INFO] [spawn_entity.py-3]: process started with pid [17154]
[gzserver-1] [1778077027.395654 [0]] gzserver: selected interface "lo" is not multicast-capable: disabling multicast
[spawn_entity.py-3] [1778077027.401053 [0]] python3: selected interface "lo" is not multicast-capable: disabling multicast
[spawn_entity.py-3] [INFO] [1778077027.447572151] [spawn_entity]: Spawn entity started
[spawn_entity.py-3] [INFO] [1778077027.447918996] [spawn_entity]: Loading entity XML from file /opt/ros/humble/share/turtlebot3_gazebo/models/turtlebot3_waffle_pi/model.sdf
[spawn_entity.py-3] [INFO] [1778077027.448685952] [spawn_entity]: Waiting for service /spawn_entity, timeout = 30
[spawn_entity.py-3] [INFO] [1778077027.448993125] [spawn_entity]: Calling service /spawn_entity
[spawn_entity.py-3] [INFO] [1778077028.204227637] [spawn_entity]: Initial orientation as reference is unset, using default value of false to comply with REP 145 (world as orientation reference)
[gzserver-1] [INFO] [1778077028.515258258] [turtlebot3_inu]: Spawn status: SpawnEntity: Successfully spawned entity [turtlebot3]
[spawn_entity.py-3] [INFO] [1778077028.628280272] [spawn_entity]: process has finished cleanly [pid 17154]
[gzserver-1] [INFO] [1778077029.023845901] [camera_driver]: Publishing camera info to [/camera/camera_info]
[gzserver-1] [INFO] [1778077029.046000733] [turtlebot3_diff_drive]: Wheel pair 1 separation set to [0.287000m]
[gzserver-1] [INFO] [1778077029.046113880] [turtlebot3_diff_drive]: Wheel pair 1 diameter set to [0.066000m]
[gzserver-1] [INFO] [1778077029.047668116] [turtlebot3_diff_drive]: Subscribed to [/cmd_vel]
[gzserver-1] [INFO] [1778077029.049879514] [turtlebot3_diff_drive]: Advertise odometry on [/odometry]
[gzserver-1] [INFO] [1778077029.052568364] [turtlebot3_diff_drive]: Publishing odometry transform between [odom] and [base_footprint]
[gzserver-1] [INFO] [1778077029.062315428] [turtlebot3_joint_state]: Going to publish joint [wheel_left_joint]
[gzserver-1] [INFO] [1778077029.062424105] [turtlebot3_joint_state]: Going to publish joint [wheel_right_joint]

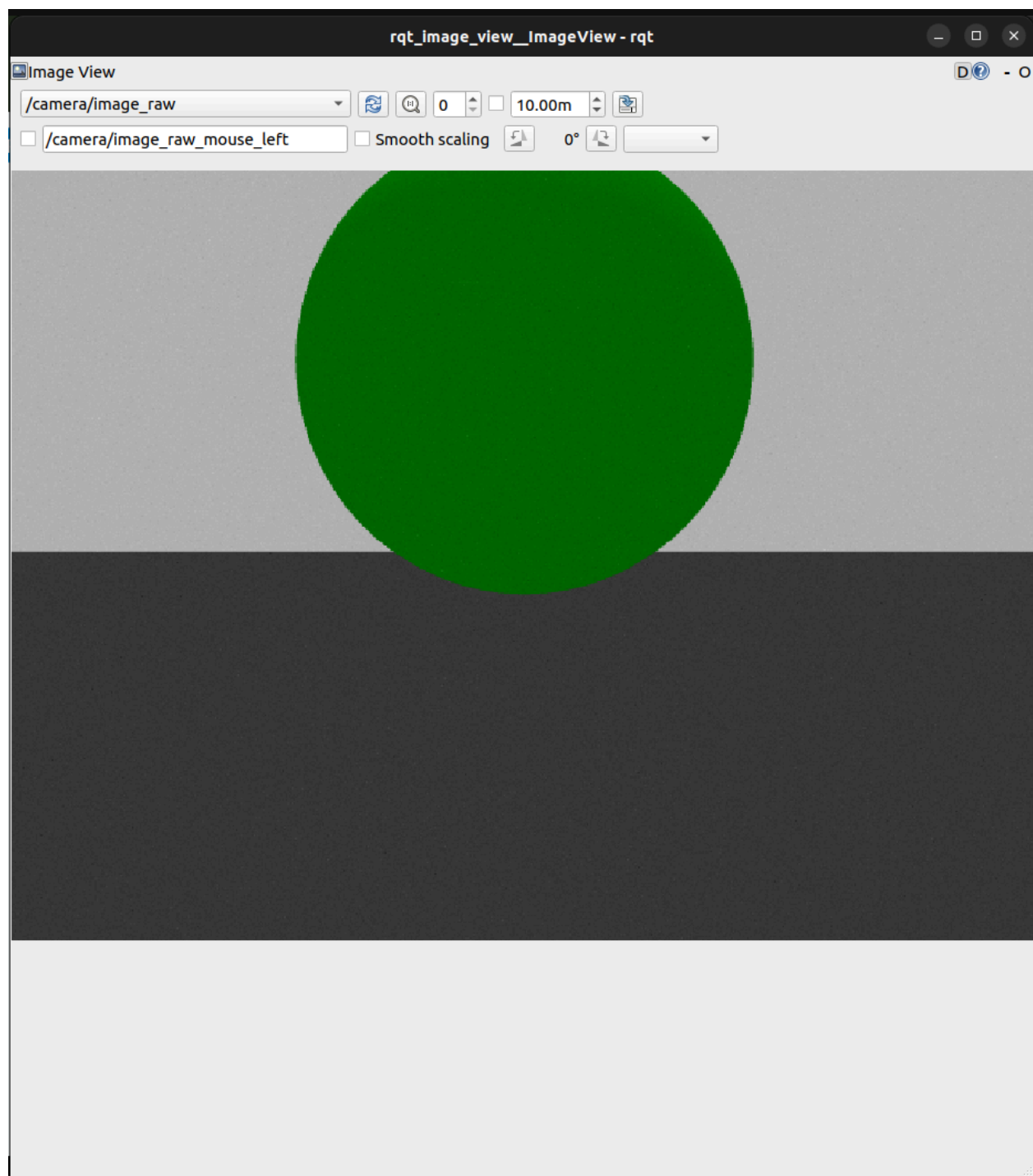
```



```
Activities Terminator May 6 19:48 priyanshu@priyanshu-hp-pavillon: ~/ros2_ws_pri

priyanshu@priyanshu-hp-pavillon:~/ros2_ws_pri$ ros2 topic echo /camera/image_raw --once
1778076917.754290 [0] ros2: selected interface "lo" is not multicast-capable: disabling multicast
header:
  stamps:
    sec: 12
    nanosec: 502000000
  frame_id: camera_rgb_optical_frame
height: 480
width: 640
encoding: rgb8
is_bigendian: 0
step: 1920
data:
- 178
- 178
- 178
- 175
- 175
- 175
- 180
- 180
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- 182
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priyanshu@priyanshu-hp-pavillon:~/ros2_ws_pri$ ros2 run rqt_image_view rqt_image_view
1778076930.539333 [0] python3: selected interface "lo" is not multicast-capable: disabling multicast
priyanshu@priyanshu-hp-pavillon:~/ros2_ws_pri$ ros2 run rqt_image_view rqt_image_view
1778077098.509725 [0] python3: selected interface "lo" is not multicast-capable: disabling multicast
```



```
Activities Terminator May 6 19:49 priyanshu@priyanshu-hp-pavilion: ~/ros2_ws_pri

[spawn_entity.py-3] [INFO] [1778077028.284727637] [spawn_entity]: Calling service /spawn_entity
[gserver-1] [INFO] [1778077028.515258258] [turtlebot3_1mu]: <Initial orientation as reference> is unset, using default value of false to comply with REP 145 (world as orientation reference)
[spawn_entity.py-3] [INFO] [1778077028.628280272] [spawn_entity]: Spawn status: SpawnEntity: Successfully spawned entity [turtlebot3]
[INFO] [spawn_entity.py-3]: process has finished cleanly [pid 17154]
[gserver-1] [INFO] [1778077029.023845901] [camera_driver]: Publishing camera info to [/camera/camera_info]
[gserver-1] [INFO] [1778077029.046000733] [turtlebot3_diff_drive]: Wheel pair 1 separation set to [0.287000m]
[gserver-1] [INFO] [1778077029.046113880] [turtlebot3_diff_drive]: Wheel pair 1 diameter set to [0.066000m]
[gserver-1] [INFO] [1778077029.047608110] [turtlebot3_diff_drive]: Subscribed to [/cmd_vel]
[gserver-1] [INFO] [1778077029.049875516] [turtlebot3_diff_drive]: Advertise odometry on [/odom]
[gserver-1] [INFO] [1778077029.052568364] [turtlebot3_diff_drive]: Publishing odom transforms between [/odom] and [/base_footprint]
[gserver-1] [INFO] [1778077029.062315428] [turtlebot3_joint_state]: Going to publish joint [wheel_left_joint]
[gserver-1] [INFO] [1778077029.062424105] [turtlebot3_joint_state]: Going to publish joint [wheel_right_joint]

...?
priyanshu@priyanshu-hp-pavilion: ~/ros2_ws_pri2$ ros2 run rqt_image_view rqt_image_view
1778076930.539333 [0] python3: selected interface "lo" is not multicast-capable: disabling multicast
priyanshu@priyanshu-hp-pavilion: ~/ros2_ws_pri2$ ros2 run rqt_image_view rqt_image_view
1778077098.599725 [0] python3: selected interface "lo" is not multicast-capable: disabling multicast

...
priyanshu@priyanshu-hp-pavilion: ~/ros2_ws_pri204x31$

[INFO] [1778077180.648451121] [color_follower]: Target Reached
[INFO] [1778077180.680237864] [color_follower]: Target Reached
[INFO] [1778077180.71148724] [color_follower]: Target Reached
[INFO] [1778077180.743830698] [color_follower]: Target Reached
[INFO] [1778077180.774819894] [color_follower]: Target Reached
[INFO] [1778077180.806949222] [color_follower]: Target Reached
[INFO] [1778077180.840699273] [color_follower]: Target Reached
[INFO] [1778077180.880876597] [color_follower]: Target Reached
[INFO] [1778077180.918655518] [color_follower]: Target Reached
[INFO] [1778077180.953977552] [color_follower]: Target Reached
[INFO] [1778077180.980224853] [color_follower]: Target Reached
[INFO] [1778077181.016161284] [color_follower]: Target Reached
[INFO] [1778077181.044602783] [color_follower]: Target Reached
[INFO] [1778077181.073883714] [color_follower]: Target Reached
[INFO] [1778077181.110418819] [color_follower]: Target Reached
[INFO] [1778077181.143343879] [color_follower]: Target Reached
[INFO] [1778077181.176935181] [color_follower]: Target Reached
[INFO] [1778077181.208427323] [color_follower]: Target Reached
[INFO] [1778077181.238324648] [color_follower]: Target Reached
[INFO] [1778077181.277254750] [color_follower]: Target Reached
[INFO] [1778077181.308685430] [color_follower]: Target Reached
[INFO] [1778077181.340218075] [color_follower]: Target Reached
[INFO] [1778077181.376448360] [color_follower]: Target Reached
[INFO] [1778077181.413808689] [color_follower]: Target Reached
[INFO] [1778077181.447615280] [color_follower]: Target Reached
[INFO] [1778077181.478263364] [color_follower]: Target Reached
[INFO] [1778077181.510226393] [color_follower]: Target Reached
[INFO] [1778077181.538077082] [color_follower]: Target Reached
[INFO] [1778077181.574197572] [color_follower]: Target Reached
[INFO] [1778077181.620307491] [color_follower]: Target Reached
```

